

Karmugilan Panchamoorthy

Research Scholar - Structural Engineering

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Profile Summary

Research scholar in Structural Engineering with research expertise in the behaviour of steel structures and cold-formed steel (CFS) shear wall systems under in-plane shear loading. Research experience spans large-scale experimental testing, finite element modelling, and experimental-numerical integration using full-scale testing and 3D Digital Image Correlation (DIC). Demonstrated capability in developing validated numerical models using Abaqus and conducting cyclic loading experiments. Strong research interest in sustainable construction practices and innovative steel structural systems.

Areas of Research Interest

- Behaviour and design of steel structures and structural connections
- Cold-Formed Steel (CFS) shear wall systems under seismic loading
- Finite Element Method (FEM) based numerical simulation (Abaqus)
- Experimental-numerical integration using full-scale testing
- 3D Digital Image Correlation (DIC) for structural assessment
- Sustainable and resilient structural systems

Core Skills

Experimental

- Large-scale and full-scale structural testing
- Cyclic and quasi-static loading protocols
- 3D Digital Image Correlation (DIC) techniques

Numerical and Computational

- Finite element modelling and analysis using Abaqus
- Nonlinear material and geometric modelling
- Model validation through experimental correlation

Programming and Software

- Programming: Python
- Structural software: STAAD
- Design standards: AISC, Eurocode, IS Codes

Professional Experience

Research Scholar (PhD)

2021 – Present

Department of Civil Engineering, IIT Hyderabad, India

Supervisor: Prof. Mahendrakumar Madhavan

- Conducted large-scale experimental testing on cold-formed steel (CFS) shear wall systems under in-plane cyclic shear loading.
- Performed full-field deformation and strain measurements using 3D Digital Image Correlation (DIC).
- Developed and validated detailed finite element models in ABAQUS for CFS shear wall behaviour.
- Integrated experimental results with numerical simulations for comprehensive structural performance assessment.
- Authored and co-authored peer-reviewed journal publications in structural engineering.

Education

Doctor of Philosophy (PhD) in Structural Engineering

2021 – Present

Indian Institute of Technology Hyderabad, India

CGPA: 9.75 / 10

Master of Engineering (M.E.) in Structural Engineering

2019 – 2021

ACGCET-Karaikudi, Tamil Nadu, India

CGPA: 9.71 / 10

Bachelor of Engineering (B.E.) in Civil Engineering

2015 – 2019

Government College of Engineering, Thanjavur (Anna University), India

CGPA: 8.44 / 10

Publications

- [1] Panchamoorthy., K. S. and Madhavan., M. M. “Behaviour of plywood-sheathed cold-formed steel shear walls under monotonic and cyclic in-plane shear loading”. In: *Journal of Constructional Steel Research* 238 (Mar. 2026). ISSN: 0143974X. DOI: [10.1016/j.jcsr.2025.110172](https://doi.org/10.1016/j.jcsr.2025.110172).
- [2] Panchamoorthy., K. S. and Madhavan., M. M. “Experimental Study of Screw Connections in Plywood-Sheathed CFS Wall Panels Subjected to In-Plane Shear Loading”. In: *Journal of Structural Engineering* 151.5 (May 2025). ISSN: 0733-9445. DOI: [10.1061/JSENDH.STENG-14106](https://doi.org/10.1061/JSENDH.STENG-14106). URL: <https://ascelibrary.org/doi/10.1061/JSENDH.STENG-14106>.

Conference Presentations

(To be added)

References

(To be added)